

2019-2020 Spring
MAT124 Midterm
08/06/2020

1. Find an equation for the line L that contains the point $P = (-1, 3, 1)$ and is orthogonal to the line

$$\frac{x-2}{-1} = \frac{y-1}{-2} = \frac{z-5}{1} = \lambda, \quad \lambda \in \mathbb{R}.$$

2. Sketch the graph of the following surfaces.

a. $z = e^y$

b. $y = z^2 - x^2$

3. The position vector for a particle in space is given as

$$\mathbf{R}(t) = (2 \cos t) \mathbf{i} + t^2 \mathbf{j} + (2 \sin t) \mathbf{k}.$$

Find the velocity and acceleration vectors of the particle and find the speed and direction of motion at $t = \pi/2$.

4. Let f be a function defined by

$$f(x, y) = \begin{cases} \frac{xy^3}{x^2 + y^6}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

Is f continuous at $(0, 0)$? Explain.

5. Use the $\epsilon - \delta$ definition and show that the function

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^6}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0), \end{cases}$$

is continuous at the origin.

6. Note that in Cartesian coordinates, the Cauchy-Riemann equations are

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

where $u = u(x, y)$ and $v = v(x, y)$.

Let $x = r \cos \theta$ and $y = r \sin \theta$. Show that the Cauchy-Riemann equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

7. According to Poiseuille's law, the resistance to the flow of blood offered by a cylindrical blood vessel of radius r and length x is

$$R(x, r) = \frac{cx}{r^4}$$

for a constant $c > 0$. A certain blood vessel in the body is 8 cm long and has a radius of 2 mm. Estimate the percent change in R when x is increased by 3% and r is decreased by 2%.

8. Find the critical points of $f(x, y) = -x^3 + 9x - 4y^2$ and classify each point as a relative maximum, a relative minimum, or a saddle point.

1. Let M be the line

$$\frac{x-2}{-1} = \frac{y-1}{-2} = \frac{z-5}{1} = \lambda, \quad \lambda \in \mathbb{R}.$$

The direction vector of M is $\mathbf{u} = \langle -1, -2, 1 \rangle$.

Let $\mathbf{v} = \langle a, b, c \rangle$, where $a, b, c \in \mathbb{R}$, be the direction vector of L . If M and L are orthogonal, the dot product of the direction vectors is zero.

$$\mathbf{u} \cdot \mathbf{v} = \langle 1, -2, 1 \rangle \cdot \langle a, b, c \rangle = a - 2b + c = 0$$

a, b, c could be any number with the relation $a - 2b + c = 0$. Let $a = 1, b = 1$. Then $c = 2b - a = 2 - 1 = 1$. The direction vector $\mathbf{v}$ is then $\mathbf{v} = \langle 1, 1, 3 \rangle$.

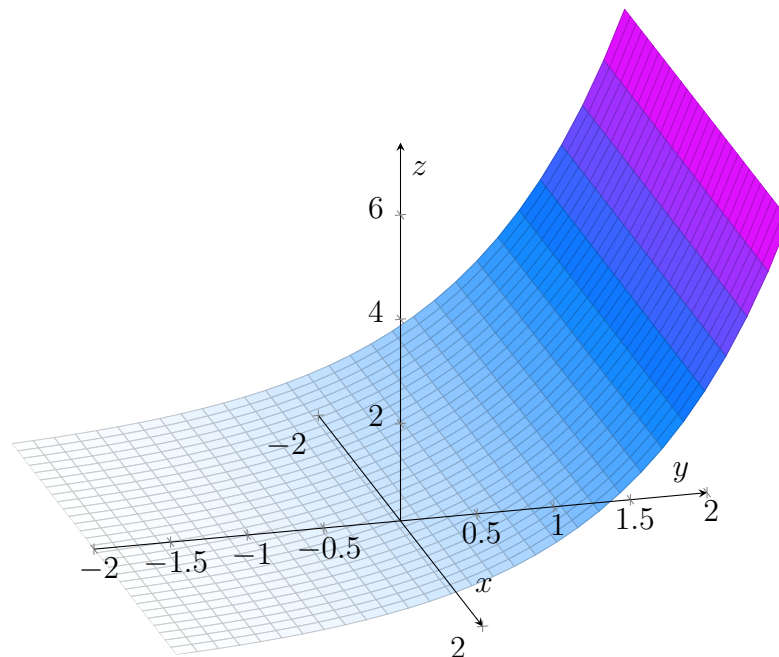
The parametric equations for a line that passes through the point $P_0(x_0, y_0, z_0)$ is given by

$$\left. \begin{aligned} x &= x_0 + v_1 t \\ y &= y_0 + v_2 t \\ z &= z_0 + v_3 t \end{aligned} \right\} \quad t \in \mathbb{R}$$

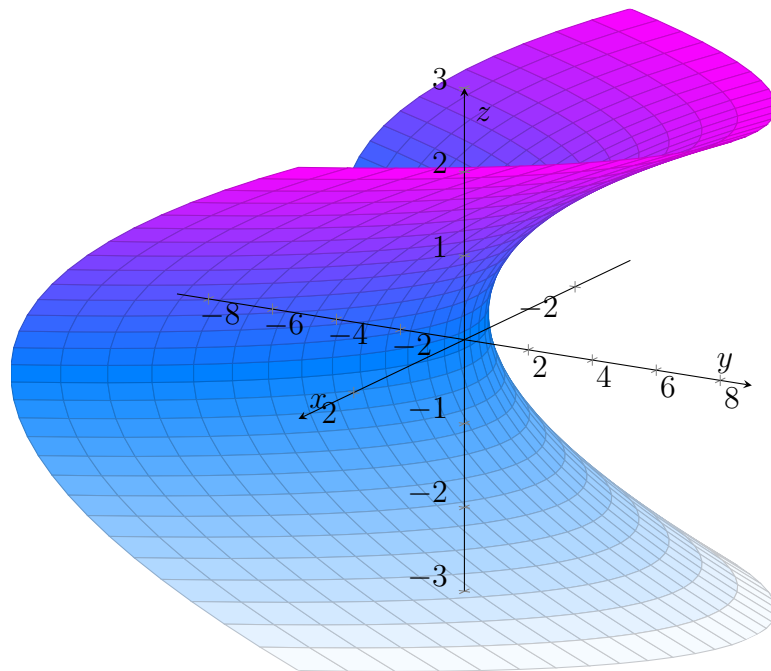
Therefore, using the point P , the parametric equations for the line L is

$$\left. \begin{aligned} x &= -1 + t \\ y &= 3 + t \\ z &= 1 + 3t \end{aligned} \right\} \quad t \in \mathbb{R}$$

2. a. Exponential surface



b. Hyperbolic paraboloid



3. The velocity and acceleration vectors can be obtained by taking the first and the second derivatives of the vector function with respect to the parametrization variable, respectively.

$$\text{Velocity vector : } \mathbf{v}(t) = \frac{d\mathbf{R}}{dt} = \langle -2 \sin t, 2t, 2 \cos t \rangle$$

$$\text{Acceleration vector : } \mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \langle -2 \cos t, 2, -2 \sin t \rangle$$

The speed of motion is the magnitude of the velocity vector, and the direction is the normalized velocity vector.

$$\text{Speed} = |\mathbf{v}(t = \pi/2)| = \sqrt{\left(-2 \sin \frac{\pi}{2}\right)^2 + \left(2 \cdot \frac{\pi}{2}\right)^2 + \left(2 \cos \frac{\pi}{2}\right)^2} = \sqrt{4 + \pi^2}$$

$$\text{Direction} = \frac{\mathbf{v}(t = \pi/2)}{|\mathbf{v}(t = \pi/2)|} = \frac{\left\langle -2 \sin \frac{\pi}{2}, 2 \cdot \frac{\pi}{2}, 2 \cos \frac{\pi}{2} \right\rangle}{\sqrt{4 + \pi^2}} = \left\langle -\frac{2}{\sqrt{4 + \pi^2}}, \frac{\pi}{\sqrt{4 + \pi^2}}, 0 \right\rangle$$

$$\mathbf{v}(t) = \langle -2 \sin t, 2t, 2 \cos t \rangle$$

$$\mathbf{a}(t) = \langle -2 \cos t, 2, -2 \sin t \rangle$$

$$\text{Speed of motion at } t = \frac{\pi}{2} : \sqrt{4 + \pi^2}$$

$$\text{Direction of motion at } t = \frac{\pi}{2} : \left\langle -\frac{2}{\sqrt{4 + \pi^2}}, \frac{\pi}{\sqrt{4 + \pi^2}}, 0 \right\rangle$$

4. Apply the Two-Path Test.

$$x = y \implies \lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{x^2 + y^6} = \lim_{y \rightarrow 0} \frac{y^4}{y^2 + y^6} = \lim_{y \rightarrow 0} \frac{y^2}{1 + y^4} = \frac{0}{1} = 0$$

$$x = y^3 \implies \lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{x^2 + y^6} = \lim_{y \rightarrow 0} \frac{y^6}{y^6 + y^6} = \lim_{y \rightarrow 0} \frac{y^6}{2y^6} = \frac{1}{2}$$

Since $0 \neq \frac{1}{2}$, by the Two-Path Test, the limit does not exist. Therefore, the function f is not continuous at $(0, 0)$.

5. The value of the function at the point $(0, 0)$ is 0. Therefore, we will show that the limit L is also 0. For every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$0 < \sqrt{(x-0)^2 + (y-0)^2} < \delta \implies |f(x, y) - L| < \epsilon.$$

$$\begin{aligned} \left| \frac{x^2 y}{x^2 + y^6} - 0 \right| &= |y| \cdot \left| \frac{x^2}{x^2 + y^6} \right| \leq |y| \cdot 1 = |y| \quad \left[\frac{x^2}{x^2 + y^6} \leq \frac{x^2}{x^2} = 1 \right] \\ &\leq |x| + |y| < 2\delta \quad \left[x^2 \geq 0, y^2 \geq 0, \sqrt{x^2 + y^2} < \delta \implies |x| < \delta, |y| < \delta \right] \end{aligned}$$

Let $\delta = \frac{\epsilon}{2}$. Then

$$\left| \frac{x^2 y}{x^2 + y^6} \right| \leq |x| + |y| = 2 \cdot \frac{\epsilon}{2} = \epsilon.$$

Since the limit is equal to the value of the function at $(0, 0)$, f is continuous at $(0, 0)$.

6. u and v are functions of x and y . x and y are functions of r and θ . Use the chain rule.

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r} = u_x \cdot \cos \theta + u_y \cdot \sin \theta = v_y \cos \theta - v_x \sin \theta$$

$$\frac{\partial v}{\partial \theta} = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial \theta} \implies \frac{1}{r} \frac{\partial v}{\partial \theta} = \frac{1}{r} (-v_x \cdot r \sin \theta + v_y \cdot r \cos \theta) = -v_x \sin \theta + v_y \cos \theta$$

$$\frac{\partial u}{\partial r} = -v_x \sin \theta + v_y \cos \theta = \frac{1}{r} \frac{\partial v}{\partial \theta}$$

$$\frac{\partial v}{\partial r} = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial r} = v_x \cdot \cos \theta + v_y \cdot \sin \theta = -u_y \cos \theta + u_x \sin \theta$$

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \theta} \implies -\frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{1}{r} (-u_x r \sin \theta + u_y r \cos \theta) = u_x \sin \theta - u_y \cos \theta$$

$$\frac{\partial v}{\partial r} = u_x \sin \theta - u_y \cos \theta = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

Therefore, the Cauchy-Riemann equations hold.

7. The total differential of R is

$$dR = \frac{\partial R}{\partial x} dx + \frac{\partial R}{\partial r} dr = \frac{c}{r^4} dx - \frac{4cx}{r^5} dr$$

Since we estimate the percent change in R , we may take $\Delta R \approx dR$. Therefore, $dx = x \cdot 3\%$, $dr = r \cdot (-2\%)$. Given also $x = 8 \text{ cm}$, $r = 2 \text{ mm} = 0.2 \text{ cm}$, the percent change can be estimated as

$$\frac{dR}{R} \cdot 100\% = \frac{1}{\frac{cx}{r^4}} \cdot \left(\frac{c}{r^4} \cdot x \cdot 3\% - \frac{4cx}{r^5} \cdot r \cdot (-2\%) \right) \cdot 100\% = \boxed{11\%}$$

8. To identify the critical points, find where both $f_x = f_y = 0$ or one of the partial derivatives does not exist.

$$f_x = -3x^2 + 9, \quad f_y = -8y$$

$$\left. \begin{array}{l} f_x = 0 \implies 9 = 3x^2 \\ f_y = 0 \implies -8y = 0 \end{array} \right\} \quad y = 0, \quad x = \pm\sqrt{3}$$

The critical points are $(\sqrt{3}, 0)$ and $(-\sqrt{3}, 0)$. To classify these points, calculate the second partial derivatives and then find the Hessian determinants.

$$f_{xx} = -6x, \quad f_{xy} = f_{yx} = 0, \quad f_{yy} = -8$$

$$(\sqrt{3}, 0) \rightarrow \left\{ \begin{array}{l} f_{xx} = -6\sqrt{3}, \quad f_{xy} = 0, \quad f_{yy} = -8 \\ \left| \begin{array}{cc} -6\sqrt{3} & 0 \\ 0 & -8 \end{array} \right| = (-6\sqrt{3}) \cdot (-8) - 0 \cdot 0 = 48\sqrt{3} > 0, \quad f_{xx} < 0 \end{array} \right.$$

$$(-\sqrt{3}, 0) \rightarrow \left\{ \begin{array}{l} f_{xx} = 6\sqrt{3}, \quad f_{xy} = 0, \quad f_{yy} = -8 \\ \left| \begin{array}{cc} 6\sqrt{3} & 0 \\ 0 & -8 \end{array} \right| = (6\sqrt{3}) \cdot (-8) - 0 \cdot 0 = -48\sqrt{3} < 0 \end{array} \right.$$

A local maximum occurs at $(\sqrt{3}, 0)$ and a saddle point occurs at $(-\sqrt{3}, 0)$.